From Detection to Interpretation

Transformer-Based Cross-Lingual Approaches to Religious Hate Speech

A *who–how–why* framework for Digital Religion research
Case: Anti-Christian discourse in Chinese, Japanese, and English

Zhidian Huang

University of Zurich · Department of Computational Linguistics

Digital Religions Conference, UZH
April 21, 2026

"How computational methods can reveal cross-cultural structures of religious hostility in digital spaces."

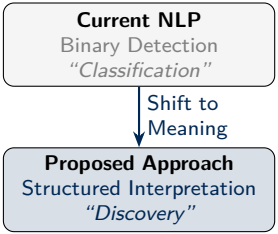
Why this talk belongs at Digital Religions

The Challenge
Digital Religion scholarship requires **scale** beyond manual reading.

The Gap
Standard NLP focuses on **Binary Detection**:
"Is this hate speech? (Yes/No)"

The Shift: Interpretive AI
Moving from labeling to **understanding**:

- **Cross-cultural narratives**
- Data-driven discovery of hostility **forms**



AI as a tool for structured interpretation.

A three-fold research gap

1. **Religion in NLP** $\approx$ Islamophobia + antisemitism.
   Anti-Christian hostility is comparatively under-studied.
2. **Languages**. Chinese and Japanese are rarely part of cross-lingual benchmarks; zero-subject grammar further breaks standard parsers.
3. **Task formulation**. Binary classification cannot tell us *who* is targeted, *how* the attack is framed, or *why* it is morally persuasive in a given community.

Takeaway

We need an *interpretable, cross-lingual* analytical framework — not a better hate-speech detector.

Why use anti-Christian discourse as the probe?

Christianity is analytically unusual because its social position **varies sharply** across the three communities we study:

- **English** — historically *dominant*, often critiqued from **within** a shared normative framework.
- **Chinese** — a *minority* faith frequently framed through **foreign influence**, missionary history, and state regulation.
- **Japanese** — a *small* minority, but politically salient after the Abe/Unification-Church scandal of 2022.

Takeaway

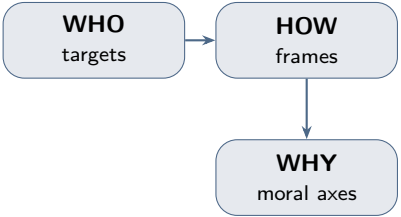
Same target, three distinct social positions — a natural laboratory for testing whether hate discourse is **universal** or **context-dependent**.

Goal: use Christianity as a **comparative probe**, not to privilege one religion.

Three linked questions

- **RQ1** — **Who** is targeted?
  Institutions, clergy, believers, public figures?
- **RQ2** — **How** is hostility expressed?
  Which rhetorical / thematic frames recur?
- **RQ3** — **Why** does it persuade locally?
  Which moral vocabularies carry the attack?

Together they form a reusable ***who–how–why* framework** for digital-religion analysis.



a reusable structured-interpretation framework

A trilingual corpus of 63,265 texts

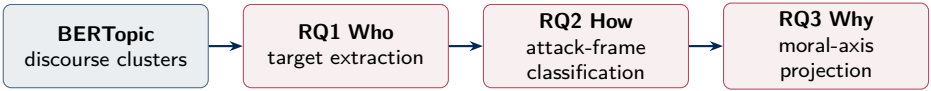
Language	Sources	Texts
English	HateXplain, Jigsaw, MLMA, Measuring Hate Speech (filtered for Christianity-targeted content)	58,940
Chinese	Baidu Tieba + Zhihu/Weibo corpora (native); LLM localisation from English seeds (synthetic)	1,965
Japanese	5ch + Common Crawl (native); LLM localisation from English seeds (synthetic)	2,360

- **Two-stage pipeline**: Stage 1 crawls native texts + trains language-specific classifiers; Stage 2 expands via **LLM-localisation** of English seeds, filtered by the same classifiers.
- Each text carries **two binary labels**: `christianity_related` & `hate_speech` — separating targeted hostility from incidental mention.

Takeaway

A balanced trilingual resource lets us compare *framings* on equal footing — and is released open-source: `github.com/notsofun/Master_Thesis`.

Pipeline: from raw text to structured meaning



Design principle.

- **Induce** structure from the corpus, don't impose an English-trained taxonomy.
- **Modular** stages — each can be inspected, challenged, or replaced by a scholar.
- **Interpretable outputs** — a target label, a frame label, a moral vector.

Why not supervised end-to-end?

- Supervised classifiers reproduce *existing* labels — mostly EN, mostly Islamophobia / antisemitism.
- Unsupervised clustering lets new structures *emerge* — e.g. the Unification-Church cluster no taxonomy pre-labelled.

Methodology: Cross-lingual Discovery & Target Extraction

Balanced Topic Modelling

- **Random sampling**: 2,000 per language ($N = 6{,}000$ total)
- BERTopic (UMAP + HDBSCAN); noise posts excluded

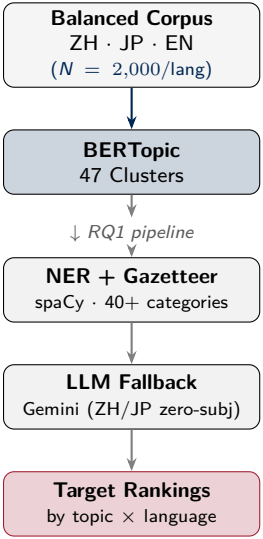
Output: **47 discourse topics** — monolingual and cross-lingual clusters coexist.

RQ1 — Target Identification

Three-stage pipeline on the topic-labelled corpus:

- **Stage 1**: spaCy NER (PERSON, ORG, NORP, GPE, FAC)
- **Stage 2**: Domain gazetteer (40+ religious categories)
- **Stage 3**: Gemini LLM fallback for ZH/JP zero-subject sentences

Output: target frequency rankings by topic × language.



Methodology: Attack frames & Moral axes

RQ2 — How? Three-layer frame extractor

1. **SVO parsing** (spaCy) — full triples where syntax permits
2. **Predicate window** ($\pm N$ words around target) — handles ZH/JP zero-subject drop
3. **LLM classification** (Gemini) — maps each predicate–argument pair to **1 of 10 attack frames** (cognitive manipulation, moral accusation, dehumanisation, …)

Taxonomy constructed inductively from 200-doc pilot; rule-based fallback ensures full corpus coverage.

RQ3 — Why? Moral-axis projection

7 axes (5 MFT + 2 intergroup-threat), constructed from multilingual moral dictionaries (MFD 2.0 / J-MFD / CMFD).

Axis vector (virtue−vice centroid in embedding space):

$$\mathbf{a}_k = \text{cent}\Big(\{E5(w)\}_{w \in D_k^+}\Big) - \text{cent}\Big(\{E5(w)\}_{w \in D_k^-}\Big)$$

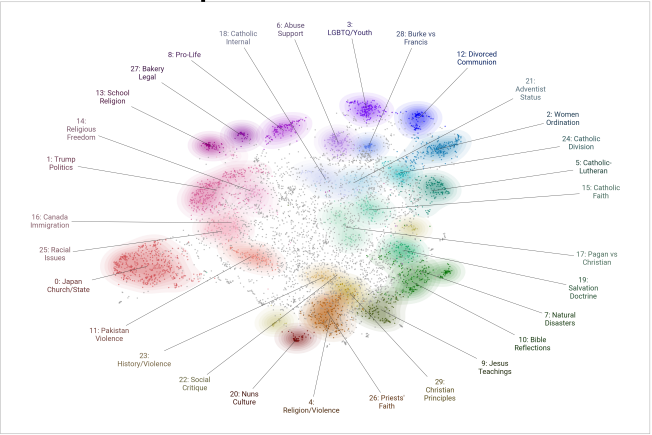
Bias score (cosine projection):

$$\text{Bias}(d, k) = \cos\big(E5(d), \mathbf{a}_k\big)$$

Negative = vice/threat pole (direct condemnation); positive = virtue pole (identity-defense grammar).

What the data says, before we ask any question

Documents and Topics



47 topics · 6,000 documents · 3 languages

Two patterns emerge:

Monolingual peaks
Events and controversies that resonate only inside one language community.

Cross-lingual clusters
Debates that attract equal participation from all three communities.

Monolingual peaks — language-specific events

JA 99% — Unification Church & Abe

Topic 0 · 499 docs · entropy ≈ 0.07

Keywords: , , ,

Example

(Abe knowingly promoted the church; the LDP let it rename itself to evade prosecution and continue "spirit sales" fraud.)

Resonates **only** in Japan — shaped by the 2022 assassination and ensuing media scrutiny.

EN 92% — Religion & Canadian Identity

Topic 16 · 61 docs · entropy ≈ 0.34

Keywords: canada, immigrants, conservative, christian, society

Example

"Canada was built on European Christian civilisational foundations; newcomers are expected to integrate into that tradition — not dilute it with incompatible values."

Religion embedded in multiculturalism debate — a framing that has **little resonance** in JA or ZH communities.

ZH 96% — Missionaries in China

Topic 43 · 22 docs · entropy ≈ 0.18

Keywords: , , , , ,

Example (ZH)

——

(Confucianism needed no missionaries — the imperial exams were its "mission structure", unlike Christianity's institutional evangelism.)

A distinctly Chinese intellectual concern with indigenous vs. Western belief systems.

Cross-lingual clusters — shared concerns across all three languages

LGBTQ+ & Catholic Teaching
Topic 3 · 149 docs · ZH 36% / EN 34% / JA 30% · *ex: ZH*

Example (ZH)

" "

(Barring gay men from the priesthood forces "pathological self-hatred"; bishops must say whether they actually enforce it.)

Universal fault line within Catholicism.

Science vs. Creationism
Topic 7 · 119 docs · EN 39% / ZH 31% / JA 30% · *ex: JA*

Example (JA)

2 2 5

(Creationists reject the definition of science: if the Bible said $2 + 2 = 5$, they'd declare mathematics a lie.)

Anti-creationist critique appears in **all three** communities.

Conservative / Liberal Church Split
Topic 36 · 25 docs · ZH 36% / EN 32% / JA 32% · *ex: EN*

Example (EN)

"Mainline denominations are getting squeezed from both ends. Conservatives call them too liberal; secular progressives have no use for faith at all. They're down to commanding maybe 17% of the American religious landscape — and shrinking."

A **global** institutional tension, not culturally specific.

Takeaway

The same critical frames appear word-for-word across ZH, EN, and JA — suggesting a shared global Catholic discourse, not three separate ones.

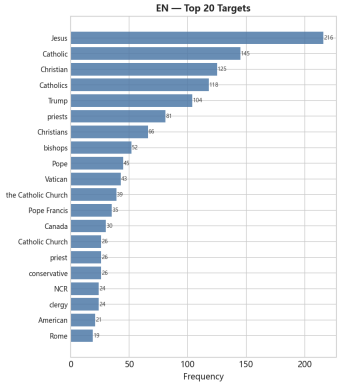
Finding 1

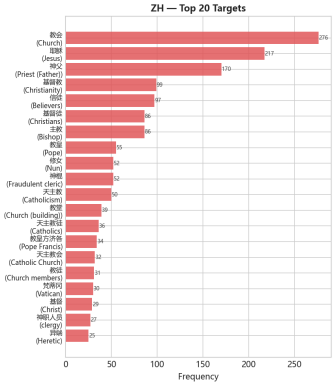
The target chooses the frame.

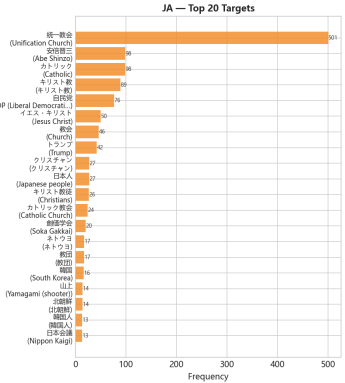
Online religious hostility follows reusable narrative templates that cross language boundaries.

RQ1 — Who is targeted?

RQ1: Global Top-20 Hate Targets by Language [原文 (EN translation)]







English
Clergy dominate — priests, bishops, Trump

Chinese
Church institutions + "religious fraudster"

Japanese
alone: 501 mentions — **politics, not theology**

RQ2 — Ten attack frames (inductively constructed)

Induced from pilot-coding 200 documents; rule-based fallback ensures full corpus coverage.

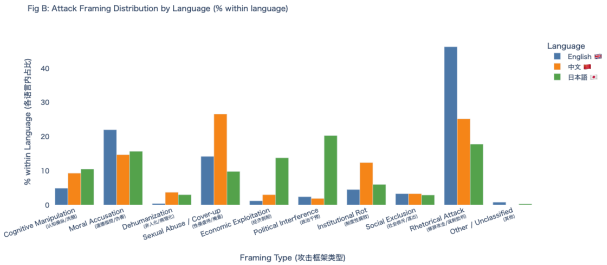
- **Rhetorical Attack** — sarcasm, mockery, invective
- **Moral Accusation** — hypocrisy, double standards
- **Sexual Abuse** — clerical abuse & cover-ups
- **Economic Exploitation** — fraud, coerced donations
- **Political Interference** — capture of state / party
- **Cognitive Manipulation** — brainwashing, cult control
- **Institutional Rot** — systemic cover-up
- **Social Exclusion** — calls for banning / ostracism
- **Dehumanization** — vermin / disease metaphors
- **Other** — residual

discourse/rhetoric institutional harm / coercion

Takeaway

Taxonomy constructed inductively — each frame maps to a distinct rhetorical move visible across all three languages.

Three national “frame signatures”



English

Rhetorical Attack 46% · Moral Accusation 22%

“The phony, fake, stifling piety.”

broad moral condemnation

Japanese

Political Interference 20% · Economic 14%

, , —

politics and money

Chinese

Sexual Abuse 26% · Rhetorical Attack 25%

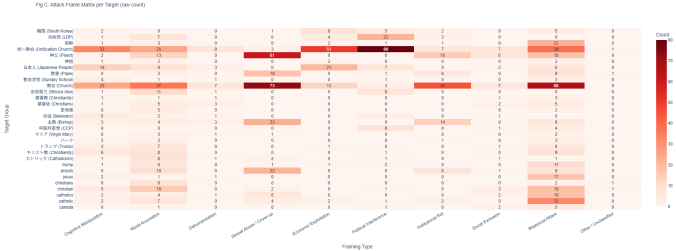
, , —

misconduct + cover-up

Takeaway

Surface vocabularies differ; the *same underlying rule* chooses them.

The stable target–frame rule



Across ***all three languages*** *— not just within one — the target category selects the frame:*

Clergy (priests, bishops, Pope) $\longrightarrow$ **Sexual Abuse Frame**
Institutions (Unification Church, LDP) $\longrightarrow$ **Political Interference / Economic Exploitation**
Collective believers (Catholics) $\longrightarrow$ **Rhetorical Attack / Moral Accusation**

Takeaway

Online hostility against religion behaves like a *genre* system: each social category of target activates a recognisable narrative template — **universal in form, localised in vocabulary**.

Finding 2

Two moral architectures, one shared target.

*English speakers attack Christianity as a **moral violator**.*
*Chinese and Japanese speakers attack it as a **threat to us**.*

How to read the moral profile

Each axis pairs a *virtue* pole with a *vice/threat* pole:

- **Harm** — care vs. harm
- **Fairness** — fairness vs. cheating
- **Loyalty** — loyalty vs. betrayal
- **Authority** — authority vs. subversion
- **Sanctity** — purity vs. degradation
- **Realistic Threat** — safety vs. invasion
- **Symbolic Threat** — cultural cohesion vs. cultural threat

Score = cosine(doc, axis) — reveals *which vocabulary is active*, not the speaker's own moral stance.

Two hostile grammars both show up:

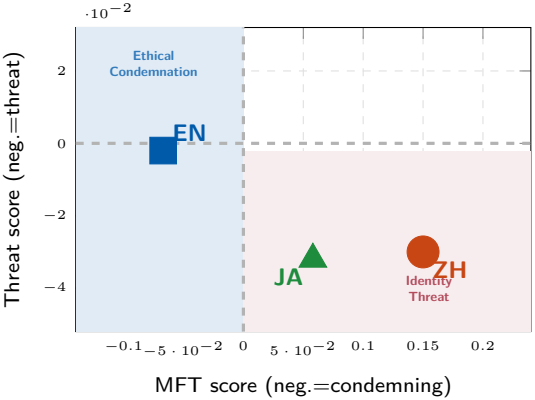
Direct condemnation

Vice words fall directly on the target.
"Christianity is corrupt, harmful, unjust."
⇒ **negative** MFT axes.

Identity defense

Ingroup virtue words defend "us"; target cast as threat.
"We must protect our sacred tradition from this foreign cult."
⇒ **positive** MFT + **negative** threat.

Three moral fingerprints



English — direct condemnation.
Negative MFT score: Christianity indicted as "harmful, unfair, corrupt, authoritarian" — a **wide ethical indictment** from within a shared normative framework.

Chinese & Japanese — identity defense.
Positive MFT axes; negative Threat axes. Christianity framed as a **foreign / social danger** — outsider critique organised around group safety.

Takeaway

Same target, two opposite moral architectures.

What the framework offers beyond this case

The *who–how–why* pipeline is **target-agnostic**. The same three-stage design can be pointed at:

- Islamophobia across Europe and South Asia
- Antisemitic rhetoric on Telegram, YouTube
- Anti-Hindu discourse in South-Asian diasporas
- Sectarian conflict (Sunni / Shi'a; Catholic / Protestant)
- Religion & nationalism studies
- Minority / majority reversals across regions

Takeaway

The contribution is a **framework**, not only a Christianity case-study — a reusable way for religion scholars to interrogate large-scale digital discourse at interpretable resolution.

Conclusions and limitations

RQ1 — Who is targeted?

Shared core (Jesus, Pope) + community-specific peaks:

JA · **ZH** clerical hierarchy **EN** clergy & politicians

RQ2 — How is hatred expressed?

Target category **selects** the frame — stably, across all three languages:

Clergy → Sexual Abuse Institutions → Political/Economic
Collectives → Rhetorical Attack

RQ3 — Why does it persuade?

EN negative on all five MFT axes *moral condemnation*
ZH JA positive MFT, negative Threat *identity defense*

Structural limits

- **Snapshot only** — cannot trace how triggering events reshape discourse over time
- **Surface antagonism** — irony, code-switching, and in-group vocabulary go undetected
- **Dictionary artefacts** — CMFD / J-MFD are imperfect adaptations of English-native MFD 2.0
- **Constructed data** — ZH / JA corpora are partly back-translation-augmented

Pipeline is **target-agnostic** and open-source —
applicable beyond the Christianity case.

Thank you.

From binary detection
toward **structured, cross-cultural interpretation**
of religious discourse online.

Zhidian Huang
huangzhidian15@gmail.com
github.com/notsofun/Master_Thesis

I'd love your questions.